

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461169.736 +/- 0.052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.055 +/- 0.04
Transit depth (Rp/Rs)^2	0.3 +/- 0.44 [%]
Orbital Inclination (inc)	89.51 +/- 2.85
Airmass coefficient 1 (a1)	1.1737 +/- 0.0068
Airmass coefficient 2 (a2)	-0.1 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.58 %
Best Comparison Star	None
Optimal Aperture	4.11
Optimal Annulus	11.33
Transit Duration (day)	0.04 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.055 ± 0.04 vs 0.1276 (deviation 0.87σ)
Duration fit vs published	0.04 d vs 0.1179 d (deviation 1.33σ)
Mid-time O–C	115.8 min
Depth SNR	0.7
Residual scatter	0.58 %

Light curve

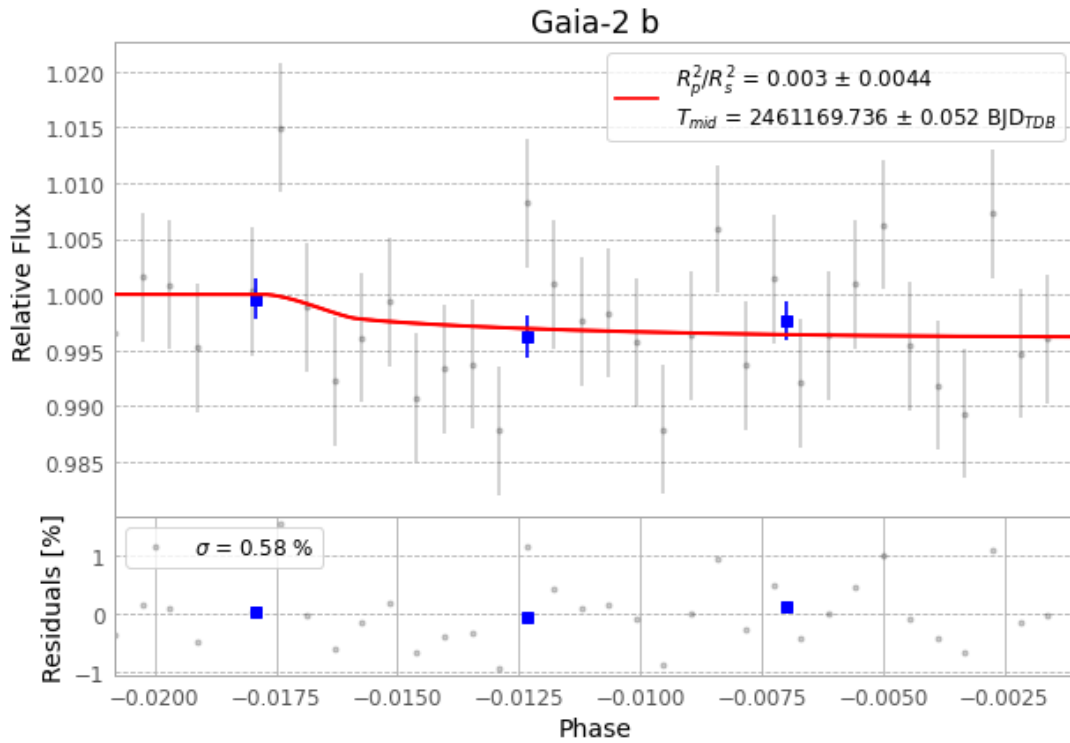


Figure 1 — FinalLightCurve_Gaia-2 b_2026-05-08.png (SHA-256 ff17eeddd335951f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1a60db2a0a9f7f94...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

36 FITS frames (24 MB), Gaia-2260509035115.FITS → Gaia-2260509053615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-260509.log (9371ace0f380...), FinalLightCurve_Gaia-2 b_2026-05-08.png (ff17eeddd335...), FinalLightCurve_Gaia-2 b_2026-05-08.csv (50ec8388fa90...)

Compute resources

Wall time 167.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 23:14Z.